Physical Gram and Moment Refinements for Exact Fourier Certificates of Order Six

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Abstract

We audit the pinned manuscript *Exact Fourier certificates for complex Hadamard matrices of order six*, separating its analytic reductions from its large rational certificates. On its precise shared-column probability space we prove an order-two function identity, a compatible physical Gram reduction, and an external-label contrast positivity constraint. We derive sharper intervals for specific original pair and mixed moments, and a lossless polyhedral account of elimination that retains the bounds on eliminated original coordinates. A rational dual witness and a residual-aware Gram inequality make these refinements replayable without assigning arbitrary bounds to new combinations. We also give a defect identity for the cubic obstruction. These are local mathematical gains, not an independent proof of the manuscript's Fourier theorem or of a new bound on mutually unbiased bases. The full reduction remains a separate verification obligation; exact block-positivity replay and the other bounded checks have only the scope stated below. Contribution assessment: C3/5, exact source-specific function compression and stronger original-moment constraints yield a notable local refinement while the full certificate and headline theorem are not independently established.

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1 Pinned Problem and Scope

The target is catalogue manuscript 537, authored by OpenAI and dated September 24, 2026, at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The source directory relative to the study's `_sources/openai-math` is

`preprints/Exact-Fourier-certificates-for-complex-Hadamard-matrices-of-order-six-September-24-2026`.

The study date is October 7, 2026. Applicability level L5 in the catalogue is a relevance assessment, not a contribution rating or verification result.

1.1 Spaces, normalization, and quantifiers

Let $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$, with conjugate-linear-first inner product on $\mathbb{C}^6$. A complex Hadamard matrix is $H \in \mathbb{T}^{6\times 6}$ satisfying $H^*H = 6I$. Since it is square, also $HH^* = 6I$. Coordinates range from 0 to 5. The charge lattice, order, and normalized character are

$$\Lambda = \{a \in \mathbb{Z}^6 : \textstyle\sum_i a_i = 0\}, \quad \operatorname{ord}(a) = \textstyle\sum_{a_i>0} a_i, \quad g_H(a) = \frac{1}{6}\sum_{k=0}^{5}\prod_i H(i,k)^{a_i}. \tag{1}$$

Thus $g_H(0) = 1$, $g_H(-a) = \overline{g_H(a)}$, and $|g_H(a)| \le 1$. Negative powers are defined because entries never vanish. Put $D = \{e_i - e_j : i \neq j\}$ and $\alpha = (1, 1, 1, -1, -1, -1)$. All row and column permutations

and all independent row and column phase multiplications are allowed in Hadamard equivalence. Balanced charges remove column phases; row phases multiply a character by the corresponding unit character. Vanishing at every coordinate permutation of α is therefore an equivalence-invariant condition.

For $\omega = \exp(2\pi i/3)$ the cubic representative $T = (\omega^{t_{ij}})$ has $t_{ij} = 0$ on its initial row, initial column, and diagonal; among indices $1, \ldots, 5$, $t_{ij} = 1$ at neighbors of a five-cycle and -1 at its other off-diagonal positions. The source's first headline is

$$\forall H \in \mathbb{T}^{6\times 6}, \quad H^*H = 6I,\ H \not\sim T \implies g_H(P\alpha) = 0 \quad \forall P \in S_6. \tag{2}$$

Its second headline excludes seven orthonormal bases $B_1, \ldots, B_7$ of $\mathbb{C}^6$ satisfying

$$|\langle b_{r,i}, b_{s,k}\rangle|^2 = 1/6 \quad \text{for every } r \neq s \text{ and } i, k \in \{0, \ldots, 5\}. \tag{3}$$

With $N(6)$ the maximal number of such bases, exclusion of seven combined with Weiner's completion theorem gives $N(6) \leq 5$, not $N(6) = 3$. The stronger companion-manuscript assertion is not an input or conclusion of this report.

There is no differential-operator boundary condition here. The physical domain is either the compact set of phase matrices obeying Hadamard equations, or the hypothetical complete MUB configuration space with (3). Every auxiliary probability space below is finite and has specified normalized counting measure. All functions lie in its entire L^2 space; there are no domain-extension, convergence, or integrability assumptions. A seven-row formal contraction is a combinatorial boundary case, not a seven-row physical sample.

1.2 What is established locally

Theorems in Sections 5–8 are proved here directly. Their hypotheses may describe a hypothetical complete family. Consequently they are necessary-constraint and certificate statements, not constructions of a seven-basis family. The cubic classification is source reasoning audited in Section 2; the rational computations completing the source dichotomy and mixed contradiction are not absorbed into our local proofs. A feasible truncated moment vector is never declared to represent a Hadamard matrix or MUB set.

2 Source-Wide Dependency Audit

The full `paper.tex`, `preamble.tex`, all six included section files, README, and bibliography were read, including the cubic sector, the finite enumeration rules, and the Appendix A wrapper and checksum contract. The appendix body is an external `lstinputlisting`; its referenced Python file was initially absent from the local acquisition, and the pinned public listing path is unresolved. The two separate standalone files, `verification/verify.py` and `verification/run.py`, were subsequently recovered from the pinned public tree by the independent auditor. The verifier's printed checksum matched, and all 576 lines, including the 1021 lower-triangle integers, were read by this author. This completes the substantive appendix reading without pretending the dangling listing path exists. Exact input hashes and read scope are recorded in the local author ledger.

2.1 Single-matrix dependencies

S1: injective moments. For a balanced integer $p \times q$ exponent matrix A with $p, q \leq 6$, the source averages

$$m_H(A) = \mathbb{E}\,\mathrm{Re}\prod_{u,v} K(\rho(u), \kappa(v))^{A_{uv}} \tag{4}$$

over $K = H, H^{\mathsf{T}}$ with weights $1/2$, and independent uniform row and column injections. Both axis sums of A vanish. Zero axes may be deleted by the uniform marginal law. Axis permutations, simultaneous negation, and transpose preserve the moment. The empty moment is 1. These facts, including $|m_H(A)| \le 1$, follow from the actual integrands, not from a formal coordinate convention.

S2: order-one contraction. If row i has a $+1$ and -1 in different columns, orthogonality of those columns, after fixing the other $p-1$ row labels, gives

$$(7-p)m_H(A) = -\sum_{\ell \neq i} m_H(A_{i\to\ell}) \quad (p \le 6). \tag{5}$$

There are exactly $7-p$ unused rows. For a formal seven-row array the six other rows exhaust the labels, so only $\sum_{\ell\neq i} m_H(A_{i\to\ell}) = 0$ remains. There is no defined $m_H(A)$ on seven injective rows. Every recursion reduces an axis length; the denominator in (5) is positive when used.

S3: equality-pattern expansion. For balanced $a_1, \dots, a_n$ with $\sum a_j = 0$, put these charges in the columns of A. If $\mathcal{P}$ partitions the factors and $A_{\mathcal{P}}$ sums the columns in each block, independent column sampling gives

$$\left\langle \operatorname{Re} \prod_{j=1}^{n} g_{\widetilde{H}}(a_j) \right\rangle_H = \sum_{|\mathcal{P}|\le 6} \frac{(6)_{|\mathcal{P}|}}{6^n} m_H(A_{\mathcal{P}}). \tag{6}$$

Here $\langle\cdot\rangle_H$ averages both orientations and all 6! row permutations with multiplicity. Each specified partition with k blocks has $(6)_k$ distinct-label assignments, proving the normalization. For $n = 2$, equal and unequal columns have weights $1/6$ and $5/6$.

S4: finite certificate, not an automatic consequence of S2. The source specifies

$$\begin{aligned} w &= (-3,-2,0,1,2,2), & a_1 &= (-1,-1,0,0,1,1),\\ a_2 &= (-2,0,0,0,1,1), & a_3 &= (-1,-1,0,1,0,1),\\ a_4 &= (-1,-1,0,1,1,0), & c &= (2,1,1,1), \end{aligned}$$

$$L_H = -g_H(w) + \sum_{j=1}^{4} c_j g_H(a_j) g_H(w - a_j).$$

For $q = (2,0,0,-2,0,0)$ and $q = (2,-2,0,0,0,0)$, separately, it claims

$$3\langle |L_{\widetilde{H}}|^2 \rangle_H + \langle |g_{\widetilde{H}}(\alpha) g_{\widetilde{H}}(q)|^2 \rangle_H = 0. \tag{7}$$

The seven-row relations selected have total order at most 11, two to four columns, positive row orders, and column orders at least two. The claimed table has 8871 nonzero equations, 909 coordinates, and 897 witness rows. These integers and the vanishing coefficient vector need an exact payload replay. Neither the set-partition identity nor the existence of true relations alone proves (7).

S5: zero-energy dichotomy. Conditional on (7), positive weight at every permutation forces each product to vanish pointwise. If one α character is nonzero, its positive and negative triples may each be reordered. The two choices of q cover cross-triple and within-triple row pairs; swapping the triples conjugates the nonzero character. Hence $g_H(2(e_i - e_j)) = 0$ for all $i \neq j$, precisely the row orthogonality of $H^{\circ 2}$. This logical inference is sound without a limiting or almost-everywhere issue because the averaging space is finite.

2.2 The entire cubic sector

S6: arbitrary phases to cube roots. For a row ratio $v \in \mathbb{T}^6$ of two different rows, both Hadamard conditions give $p_1(v) = p_2(v) = 0$. Newton's identities give $e_1 = e_2 = 0$; unimodularity gives $e_{6-k} = e_6\overline{e_k}$, hence $e_4 = e_5 = 0$. Its root polynomial is $z^6 - e_3 z^3 + e_6$, so the six entries form two rotated cube-root triples, possibly coincident. The same statement applies to columns.

After dephasing, each row of $H^{\circ 3}$ is constant 1, or consists of three 1's and three copies of $x \neq 1$. Nonconstant supports are subsets of size three of the five noninitial positions. Distinct supports have intersection $k = 1$ or 2. For values x, y, their cubed ratio has values $1, x, y^{-1}, xy^{-1}$ with multiplicities $k, 3-k, 3-k, k$. At most two values are allowed, forcing $x = y^{-1}$ and $x^2 \in \{1, x\}$, thus $x = y = -1$. That would give multiplicities 2 and 4, contradicting 3 and 3. The supports therefore coincide, for both rows and columns. A nontrivial cubed matrix must have, after ordering, the block form

$$H^{\circ 3} = \begin{pmatrix} J_3 & J_3 \\ J_3 & xJ_3 \end{pmatrix}, \quad x \neq 1. \tag{8}$$

Restrict the two classes of rows to the last three columns. Opposite classes have orthogonal restrictions because each separated rotated triple sums to zero. Each class spans at least two dimensions: if three restrictions spanned a line, full-row orthogonality would force equality in Cauchy–Schwarz for their complementary restrictions, putting three orthogonal full rows in a space of dimension at most two. Two mutually orthogonal subspaces of dimension at least two cannot lie in $\mathbb{C}^3$. Thus (8) is impossible and the dephased entries are cubic.

S7: cubic classification. Each noninitial row and column contains each cube root twice. The extra ones form a matching and are put on the diagonal. On the other positions write exponents $s_{ij} = \pm 1$. Every row has two plus signs. Orthogonality of rows i, j gives exactly two sign mismatches among the three positions outside $0, i, j$, so their product is 1. The product of each row's four signs is also 1, yielding $s_{ij} = s_{ji}$. The plus graph is a simple 2-regular graph on five vertices and therefore one five-cycle. This is T. This finite graph inference does not classify general order-six Hadamard matrices; the entrywise-square premise is essential.

S8: symmetry and the complete-family cubic obstruction. The source constructs $36 \cdot 2 \cdot 10 = 720$ normalized symmetries by choosing an initial row and column, conjugation, and cycle ordering. In exponent columns $X = \{t_0, \ldots, t_5\} \subset (\mathbb{Z}/3)^6$, a symmetry fixing row order acts as $sX + u$, $s = \pm 1$. No nonzero translation preserves X since it would put both t_j and $-t_j$ in X. Negation does not preserve X either. For $s = -1, u = t_x$, choose a neighbor y and common non-neighbor z; $t_x - t_y$ has its only noninitial zero at z but disagrees with t_z in positions x, y. Thus the 720 operations have distinct row permutations. This licenses the covariance extension from representative cubic counts to full signed permutation orbits; counting representatives alone would not.

Fixing a cubic transition, let $G = g_T$ and let f average the other five character functions. Completeness gives $f = -G/5$ at order two, and cross-unbiasedness gives

$$\sum_{d \in D} f(a+d)\overline{G(c+d)} = 0 \quad (a, c \in \Lambda). \tag{9}$$

The source's 11 orbit representatives, cube-root coefficient counts, and two 30-shift counts yield

$$S_6 = S_8 = -1/20, \quad S_3 + S_7 = 1/5, \quad S_0 + 12S_3 = 2/5, \quad 4S_3 + 2S_7 = 3/5. \tag{10}$$

Here $S_j = \mathbb{E}_{P\in S_6} \operatorname{Re}(f(Pw_j)\overline{G(Pw_j)})$ in the source's table order. They force $S_3 = S_7 = 1/10$, $S_0 = -4/5$. But $w_0 = 3(e_5 - e_0)$ and a sampled other-basis column z obey

$$0 \le \mathbb{E}\left|\sum_i z_i^3\right|^2 = 6 + 30S_0 = -18. \tag{11}$$

This is the source's complete-family cubic exclusion, not its stronger literature antecedent excluding a third basis. Section 9 extracts an exact local error account from these equations.

2.3 Complete-family and mixed dependencies

For fixed reference r, $H_{rs}(i,k) = \sqrt{6}\langle b_{r,i}, b_{s,k}\rangle$. The seven traceless diagonal matrix spaces are mutually orthogonal of dimension five and fill the 35-dimensional traceless space. Orthogonal projection, paired bilinearly, gives

$$\operatorname{tr}(XY) = \sum_{s=1}^{7}\sum_{k=0}^{5} \operatorname{tr}(XP_{s,k})\operatorname{tr}(YP_{s,k}), \qquad \operatorname{tr} X = \operatorname{tr} Y = 0. \tag{12}$$

Using two off-diagonal matrix units of noncanceling total charge gives $\sum_{s\neq r} g_{H_{rs}}(a) = 0$ for every order-two a. This is where completeness, rather than merely a Hadamard or MUB pair, enters. Column orthogonality and pairwise unbiasedness separately give

$$\sum_{j=0}^{5} g_s(a + e_i - e_j)g_s(b - e_i + e_j) = g_s(a+b), \tag{13}$$

$$\sum_{d\in D} g_s(a+d)g_t(b-d) = 0 \quad (s \neq t). \tag{14}$$

The first follows because the two column samples must coincide after orthogonality; the second uses $|\sum_i x_i/y_i|^2 = 6$ for columns from different transitions. Reversing both shifts is also allowed in (13).

The source's mixed reductions use precisely these identities, common coordinate permutations, simultaneous charge negation, label renaming, zero-factor removal, and the all-pairs α vanishing premise. An order-two singleton label among l used labels has $7 - l$ available external bases, so its replacement factor is $-1/(7-l)$. The integer l is at most six, hence this denominator does not vanish. Single-label keys use (6); exchanging r, s sends H_{rs} to $\overline{H_{rs}}^{\mathsf{T}}$, and the real part removes conjugation, justifying transpose averaging for pair moments.

The mixed enumeration restricts balanced triples to summed order at most 11 and individual seed orders at most 5; shifted equations are included only within the stated summed-order limit. Selecting finitely many valid equations does not require proving completeness of that list. The claimed mixed table has 46041 nonzero equations, 3429 coordinates, and 2767 witness rows. Six real symmetric integer blocks have sizes $5, 21, 11, 27, 22, 12$ (98 total) and decimal exponents $14, 13, 13, 14, 14, 15$. The order of the two Young permutation actions is $(P_pP_qv)_i = v_{q(p(i))}$, so reversing them changes the certificate.

After exact elimination the source claims for its nonnegative form $\mathcal{Q} = 5\sum_\lambda 10^{15-e_\lambda}\mathbb{E}(Y_\lambda^* Z_\lambda Y_\lambda)$ the coefficients

$$c_0 = -\frac{613302797399911}{6480}, \quad \sum_{K\neq 0}|c_K| = \frac{179721388988719}{6480}, \quad c_0 + \sum_{K\neq 0}|c_K| = -\frac{2007321335237}{30}. \tag{15}$$

The normalized form divides these by $5 \cdot 10^{15}$, giving negative gap magnitude $\frac{2007321335237}{150000000000000000}$. Recomputing this subtraction is not recomputing the coefficient vector. The full payload must certify every pivot, every identity, every named residual coordinate, and these totals. The reported pivots exceeding 40000 prove positivity through Schur complements but do not by themselves give a smallest-eigenvalue bound of 40000.

3 Actual Private Corpus Correspondences

The following paths are local confidential sources, not public references. The passages, rather than routing maps, were read. Their exact locations are recorded here so that the representation choices can be audited.

3.1 Positive measures and zero energy

`C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch02/finite.tex`

The sections "What a Moment Record Specifies" and "Finite Atomic Recovery With a Complete Check", labels `v21c02:hankel`, `v21c02:localizer`, `v21c02:zero-energy`, and `v21c02:atomic-certificate`, separate positivity, support, existence, and identification. Their $\int |p|^2 \, d\mu$ identity transfers here by replacing monomials on a real line by the concrete functions $F_t(a)$ on a finite sampling space. No univariate Hankel characterization is applied to complex phase points. The exact zero-energy principle accounts for source step S5. Atomic recovery is not invoked: our atoms are already the finite sample labels, and no unknown measure is extracted from a truncated Gram matrix.

3.2 The physical metric and its nullspace

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The sections "A Nonorthogonal Representation Carries a Metric", "The Full Residual Has an Omitted Component", and "Approximating the Metric", at labels `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, `v21c01:residual-decomposition`, `v21c01:residual-identity`, and `v21c01:gram-equivalence`, require the Gram metric J^*J and retain omitted residuals. For our sampling synthesis $Jc = \sum c_j \psi_j$, J need not be injective: order-two completeness creates an exact kernel. Accordingly J^*J is initially semidefinite, and it is positive definite only after a justified quotient by the actual kernel. We never invert a singular sample Gram or infer completeness from a finite projection.

3.3 Analysis, synthesis, and residual certificates

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The subsections "What a Ram representation means in this report" (`eq:moments`), "Ram moment coordinates that retain the complex" (`thm:transfer`), and "A concrete polynomial Ram construction" define a finite measurement bank $m = Ec$, its compatible inverse, and metric transport. In this report a Ram representation means exactly such a finite analysis/synthesis description of the actual Fourier test functions and their physical L^2 metric. It is not a claim that changing coordinates changes the admissible Hadamard set.

"Global Optimality Sandwich", theorem "Replayable global gap certificate" (eq:gap), and "Residual-Aware SOS Lower Bounds", theorem "Inexact Gram and identity certificate" (sec:soserror047, eq:gramerror047), require exact coefficient comparison or explicit residual charges and quantified Gram defects. Our Section 8 specializes this account to bounded shared-column functions, using their sharper type-dependent amplitudes. The classical ingredients are Gram positivity, conditional expectation, congruence, finite-dimensional elimination, and linear-program duality. The private Ram contribution is their explicitly matched representation and error account. Historical priority for these classical techniques is neither reassigned nor claimed.

4 Shared-Column Physical Gram Space

Assume for this section that a complete family satisfies (3). Sample r uniformly from seven bases, $\sigma \in S_6$ uniformly, and a bijection $\pi : \{0,\dots,5\} \to \{1,\dots,7\} \setminus \{r\}$ uniformly and independently conditional on r. Finally sample a column $k \in \{0,\dots,5\}$ uniformly and set $z = H_0(:,k)$, where

$$H_s(i,k) = \sqrt{6}\langle b_{r,\sigma(i)}, b_{\pi(s),k}\rangle.$$

The sample space Ω has $7(6!)^2 6$ equiprobable labels. Equal numerical outcomes are retained with multiplicity. Let $\mathcal{H} = L^2(\Omega,\nu;\mathbb{C})$ with $\langle u,v\rangle = \mathbb{E}\overline{u}v$. For $a \in \Lambda$ define

$$A_s(a) = g_s(a)z^{-a}, \quad F_0(a) = A_0(a), \quad F_1(a) = \frac{1}{5}\sum_{s=1}^{5} A_s(a). \tag{16}$$

Every occurrence uses the same z. These are bounded measurable functions, with $|A_s(a)| \le 1$ pointwise. Write

$$\mathcal{M}(\mathbf{s};\mathbf{a}) = \mathbb{E}\,\mathrm{Re}\prod_j g_{s_j}(a_j), \qquad \sum_j a_j = 0.$$

The expectation for $\mathcal{M}$ does not need k, since its integrand is independent of k. Distinct labels have the uniform injective marginal.

Proposition 4.1 (The shared-column Gram identity)**.** *For balanced a, v put $b = (a, -v, v-a)$. Then*

$$\mathrm{Re}\,\mathbb{E}F_s(a)\overline{F_t(v)} = \mathcal{M}((s,t,0);b), \quad (s,t) \ne (1,1), \tag{17}$$

$$\mathrm{Re}\,\mathbb{E}F_1(a)\overline{F_1(v)} = \tfrac{1}{5}\mathcal{M}((1,1,0);b) + \tfrac{4}{5}\mathcal{M}((1,2,0);b). \tag{18}$$

For any finite family $\psi_j = F_{t_j}(a_j)$, its Hermitian physical Gram $G_{ij} = \mathbb{E}\overline{\psi_i}\psi_j$ is positive semidefinite. Its real part is a positive semidefinite real matrix, and the displayed formulas compute it (with conjugate convention immaterial after real part).

Proof. Condition on r, σ, π. The column average gives $\mathbb{E}_k z^{v-a} = g_0(v-a)$. With one external average all five choices have identical expectations. With two averages there are 5 equal-label and 20 unequal-label choices out of 25. This proves both identities. For every $c \in \mathbb{C}^n$, $c^*Gc = \mathbb{E}|\sum c_j\psi_j|^2 \ge 0$. For real c, this is $c^{\mathsf{T}}(\mathrm{Re}\,G)c$; hence the real Gram is positive semidefinite. Independent columns for different functions would not give $g_0(v-a)$ and would not compute this Gram. □

5 Order-Two Compression and Contrast Positivity

Theorem 5.1 (Pointwise order-two compression)**.** *Under the complete-family premise, for every $a \in \Lambda$ of order two,*

$$F_1(a) = -\tfrac{1}{5} F_0(a) \quad \text{at every sample in } \Omega. \tag{19}$$

For order-one charges both functions are zero; for charge zero both are one. If every transition also has vanishing permuted α characters, both functions are zero at these charges. These identities remain valid after any of the source's signed Young permutation sums.

Proof. Fix the sample before selecting k. Equation (12) gives $g_0(a) + \sum_{s=1}^{5} g_s(a) = 0$ at order two. Multiplying by z^{-a} proves (19) pointwise, not merely after averaging. The other assertions use orthogonality, $g_s(0) = 1$, or the stated extra vanishing premise. Coordinate permutations preserve each of these charge classes. Apply the pointwise identity term by term in the finite signed sum. There is no assertion for a general order-three charge. □

Corollary 5.2 (Exact transport of certificate blocks)**.** *Expand a finite source function vector Y into its signed charges and apply Theorem 5.1. Collect equal remaining functions in a dictionary V. There is an explicitly rational matrix S with $Y = SV$ pointwise. For each real symmetric positive semidefinite Z,*

$$\mathbb{E} Y^* Z Y = \mathbb{E} V^* (S^* Z S) V, \qquad S^* Z S \succeq 0. \tag{20}$$

No independence of the dictionary is required for this equality. The coefficient Gram of the function synthesis is transported by congruence under any injective reparametrization, and is positive definite only on the quotient by the actual zero-function kernel.

Proof. All replacements have coefficients $0, 1, -1/5$ and the Young coefficients are integers. Summation and collection give S over $\mathbb{Q}$. Substitution proves the identity. For every vector c, $c^* S^* Z S c = (Sc)^* Z (Sc) \geq 0$. For synthesis J, $\ker(J^*J) = \ker J$ since $c^* J^* J c = \|Jc\|^2$. This proves the final assertion. □

This removes an exact source-specific redundancy whenever such order-two type-1 functions are present. A bounded inspection of the actual 98 source entries finds *none*: all 59 type-1 entries have order three, and all 14 order-two entries have type zero. Thus this particular identity removes zero entries from the supplied lists. Its realized result is an exact representation theorem for extensions and an independent consistency constraint, not an observed compression of these six blocks. No speedup is reported. A coordinate congruence alone does not tighten an exact certificate; the accompanying extra interval and contrast constraints are separate mathematical information. For specificity, the exact literal inventory is as follows. There are no zero, order-one, or permuted-α charges in any of the six lists; the same remains true in their Young sums because permutation preserves these properties.

Shape	Type 0, order 2	Type 0, order 3	Type 1, order 3	Total
$(1,1,1,2)$	0	2	3	5
$(1,1,3)$	2	6	13	21
$(1,2,2)$	2	3	6	11
$(1,4)$	4	6	17	27
$(2,3)$	4	5	13	22
(5)	2	3	7	12
Total	14	25	59	98

Thus neither the zero rules nor the order-two type identification alone reduces the size of an actual supplied block. We did not test for all other possible physical linear dependencies, which could require additional moment relations. These counts prevent a general compression principle from being mistaken for an executed block reduction.

Theorem 5.3 (External-label contrast Gram)**.** *Choose finitely many charges $a_1, \dots, a_n$. Define Hermitian matrices*

$$D_{ij} = \mathbb{E}\overline{A_1(a_i)}A_1(a_j), \qquad S_{ij} = \mathbb{E}\overline{A_1(a_i)}A_2(a_j).$$

Then

$$D - S \succeq 0, \qquad D + 4S \succeq 0, \qquad G_{F_1} = \tfrac{1}{5}(D + 4S). \tag{21}$$

If every listed charge has order two, and G_0 is the Gram of $(F_0(a_i))_i$, then

$$D + 4S = \tfrac{1}{5}G_0, \qquad S = \tfrac{1}{20}G_0 - \tfrac{1}{4}D. \tag{22}$$

Proof. Permuting labels $1, \dots, 5$ leaves the law unchanged while fixing the sampled label 0 and its column. Thus every diagonal label block in the Gram of all $5n$ functions is D and every off-diagonal block is S. Exchanging labels 1 and 2 proves $S = S^*$. The full Gram is $I_5 \otimes (D - S) + \mathbf{1}\mathbf{1}^{\mathsf{T}} \otimes S$. Restricting label coefficients to $\mathbf{1}^{\perp}$ gives $D - S \succeq 0$, and to $\operatorname{ran}\mathbf{1}$ gives $D + 4S \succeq 0$. Averaging all 25 blocks proves the last identity in (21). At order two Theorem 5.1 gives $G_{F_1} = G_0/25$, proving (22). □

The source's F_1 uses the average sector. The contrast sector supplies additional necessary Gram inequalities with the same shared-column three-factor expansion. A future larger relaxation can retain it; no numerical improvement of the source's six-block certificate is asserted.

6 Sharper Bounds on Named Original Moments

Let $m(A) = 42^{-1}\sum_{r\neq s} m_{H_{rs}}(A)$ when a complete family is being discussed. The statements about a single m_H need only one Hadamard matrix. All bounds in this section attach to specified original moments, not to linear combinations introduced by elimination.

Theorem 6.1 (Pair and mixed interval refinement)**.** *For every balanced charge a,*

$$-\tfrac{1}{5} \le m_H([a, -a]) \le 1. \tag{23}$$

The same holds for the ordered-pair average m. If a has order one, then $m_H([a, -a]) = -1/5$; the same equality holds when every permuted character of a vanishes for both orientations. For a complete family, put

$$T_a = \mathcal{M}((0, 0); (a, -a)), \qquad U_a = \mathcal{M}((0, 1); (a, -a)).$$

Then $0 \le T_a \le 1$, and for every order-two a,

$$T_a = \frac{1 + 5m([a, -a])}{6}, \qquad U_a = -\tfrac{1}{5}T_a = -\tfrac{1}{30} - \tfrac{1}{6}m([a, -a]), \qquad -\tfrac{1}{5} \le U_a \le 0. \tag{24}$$

Proof. The $n = 2$ case of (6) is $\langle |g_{\widetilde{H}}(a)|^2\rangle_H = (1 + 5m_H([a, -a]))/6$. Its left side lies in $[0, 1]$, proving (23). If that left side vanishes, its right side gives $-1/5$. At order one both orientations have the required orthogonality. The ordered-pair law is transpose invariant after conjugation, as established in the source audit; hence the same expansion gives $T_a = (1 + 5m([a, -a]))/6$. At order two completeness gives $g_0(a)\sum_{t=0}^{5}\overline{g_t(a)} = 0$. Taking real expectations and using exchangeability among the five nonzero labels gives $T_a + 5U_a = 0$. The displayed affine formula and interval follow. □

The nonnegative same-label square bound holds at every charge, not just order two. The vanishing conclusion for m_H must include transpose orientation; vanishing for H alone at a fixed non-symmetric charge does not silently establish that premise for H^{T}.

Corollary 6.2 (Certified interval support bound)**.** *Suppose an exact expansion is* $\mathcal{Q} = c_0 + \sum_{i=1}^{N} c_i \mu_i$ *and the original named moments have proved intervals* $[\ell_i, u_i]$*. Then*

$$\mathcal{Q} \le c_0 + \sum_i \max\{c_i \ell_i, c_i u_i\}. \tag{25}$$

Replacing $[-1, 1]$ *by any of the contained intervals in Theorem 6.1 cannot increase this upper bound. A pair coordinate in (23) with* $c_i < 0$ *lowers its old contribution from* $|c_i|$ *to* $|c_i|/5$*; an order-two distinct-label coordinate with* $c_i > 0$ *in (24) contributes at most 0.*

Proof. The maximum of a scalar linear function on a closed interval is attained at an endpoint. Sum the individual bounds. Shrinking the interval cannot increase its support function; the two stated values follow directly. □

An original residual key may or may not have these forms after canonical renaming. The source's residual coefficient list is needed before a numerical improvement can be attributed to these refinements.

7 Lossless Elimination and Rational Dual Witnesses

Let $\mu \in \mathbb{R}^N$ include its constant moment $\mu_0 = 1$. Let $W \in \mathbb{Q}^{r \times N}$ consist of valid homogeneous equations. Select r columns J such that W_J is invertible, with complementary indices F. Set $A = -W_J^{-1} W_F$. The source certifies invertibility by a nonzero determinant modulo a prime, then verifies a rational substitution P by $WP = 0$, identity free rows, and zero pivot columns. In the (J, F) order this is $P = \left(\begin{smallmatrix} 0 & A \\ 0 & I \end{smallmatrix}\right)$.

Theorem 7.1 (Original-coordinate polytope after elimination)**.** *Give every original coordinate an interval* $\ell_i \le \mu_i \le u_i$*, with* $\ell_0 = u_0 = 1$*. Then the map*

$$x \longmapsto \mu = (Ax, x) \tag{26}$$

is a bijection from

$$\mathcal{B} = \{x : \ell_F \le x \le u_F, \quad \ell_J \le Ax \le u_J\} \tag{27}$$

onto the original bounded equation set $\{\mu : W\mu = 0,\ \ell \le \mu \le u\}$*. Therefore for any rational coefficient vector* c*,*

$$\sup_{\mu} c^{\mathsf{T}} \mu = \sup_{x \in \mathcal{B}} (c_F + A^{\mathsf{T}} c_J)^{\mathsf{T}} x. \tag{28}$$

If nonempty, these polytopes are compact and both suprema are attained. Dropping the inequalities on Ax *is a valid relaxation but can strictly weaken the bound.*

Proof. $W\mu = 0$ is exactly $W_J \mu_J + W_F \mu_F = 0$, hence $\mu_J = A\mu_F$. This proves bijectivity and transports every interval without altering its coordinate meaning. Substitution proves the objective identity. The original product of finite closed intervals is compact; the equation set is closed. Equivalently $\mathcal{B}$ is closed and bounded by its free-coordinate intervals. Dropping constraints enlarges it. □

This distinguishes two operations often conflated in certificate replay. A pivot coordinate expressed as a combination is still an original bounded observable, so its interval must be imposed on that combination if a lossless bounded model is intended. Conversely a newly defined combination is not automatically bounded by one. The source uses the weaker free-coordinate box legitimately; it does not assert the stronger false boundedness statement.

Proposition 7.2 (Finite rational witness without LP trust)**.** *For any rational vector $y \in \mathbb{Q}^r$, put $d = c - W^{\mathsf{T}} y$. Every original moment vector satisfying $W\mu = 0$ and the given intervals obeys*

$$c^{\mathsf{T}}\mu \le h_{[\ell,u]}(d) := \sum_{i=0}^{N-1} \max\{d_i\ell_i, d_i u_i\}. \tag{29}$$

The constant contributes exactly d_0. If $|(W\mu)_j| \le \delta_j$ instead, the valid right side is

$$h_{[\ell,u]}(c - W^{\mathsf{T}} y) + \sum_j |y_j|\delta_j. \tag{30}$$

All inequalities are verified by rational multiplication and comparisons when the data and error bounds are rational. A search algorithm need not be trusted, and an optimal dual vector is not required.

Proof. Write $c^{\mathsf{T}}\mu = d^{\mathsf{T}}\mu + y^{\mathsf{T}}W\mu$. Bound each $d_i\mu_i$ on its interval and the residual term by $\sum |y_j|\delta_j$. This also proves the exact case. □

Finite-dimensional LP duality relates the infimum of these bounds to (28) when the original bounded equation set is nonempty. The displayed weak-duality proof suffices for every witness used here and continues to be sound in an infeasibility argument; no strong-duality assumption is needed to conclude that a negative upper bound contradicts an independently nonnegative form.

7.1 Strict improvement in a transparent rational example

This example isolates the elimination issue, not an asserted source residual. Let $\mu = (1, x, y)$, $x - 2y = 0$, and $|x|, |y| \le 1$. Eliminating x and using only $|y| \le 1$ bounds the objective y by 1. The original bound $|x| \le 1$ becomes $|2y| \le 1$, so the exact bound is $1/2$, attained at $(x, y) = (1, 1/2)$. In Proposition 7.2, choose $c = (0, 0, 1)$, $W = (0, 1, -2)$, and multiplier $-1/2$; then $c - W^{\mathsf{T}}(-1/2) = (0, 1/2, 0)$ and the upper bound is $1/2$. The combination $2x + y$ at that feasible point is $5/2$. Assigning it a unit bound would be false. This example proves a possible strict gain from the lossless polytope, but is not measured performance on M537.

Lemma 7.3 (Exact residual transport)**.** *For arbitrary real μ, the exact substitution matrix satisfies*

$$\mu - P\mu = \iota_J W_J^{-1} W\mu, \tag{31}$$

where ι_J inserts a vector into the pivot coordinates. Thus

$$|c^{\mathsf{T}}(\mu - P\mu)| \le \sum_j |(c_J^{\mathsf{T}} W_J^{-1})_j|\delta_j \quad \text{if } |(W\mu)_j| \le \delta_j.$$

Proof. The free coordinates of $\mu - P\mu$ vanish. Its pivot coordinates are $\mu_J + W_J^{-1}W_F\mu_F = W_J^{-1}W\mu$. Multiply by c and use the componentwise residual bounds. Rational exactness of P does not require rationality of the physical moments. □

8 Residual-Aware Physical Certificates

The source has integer blocks and claims exact arithmetic. The following extension states precisely what would remain valid if proposed blocks or expansions had certified defects. It neither substitutes floating point for its exact contract nor asserts an approximate-MUB theorem.

8.1 Amplitude bounds derived from the actual functions

Under the all-pairs vanishing premise define, for each type and charge,

$$b_t(a) = \begin{cases} 0, & \operatorname{ord}(a) = 1 \text{ or } a \in S_6\alpha, \\ 1/5, & t = 1,\ \operatorname{ord}(a) = 2, \\ 1, & \text{otherwise.} \end{cases} \tag{32}$$

If the vanishing premise is absent, remove only the $S_6\alpha$ case. Theorem 5.1 proves $|F_t(a)| \le b_t(a)$ pointwise. If a Young vector, after collecting equal charges, has

$$(Y_\lambda)_i = \sum_{t,a} q_{\lambda i,t,a} F_t(a), \quad q_{\lambda i,t,a} \in \mathbb{Q},$$

then the explicit rational bound

$$\mathbb{E}\|Y_\lambda\|_2^2 \le B_\lambda := \sum_i \left(\sum_{t,a} |q_{\lambda i,t,a}| b_t(a) \right)^2 \tag{33}$$

follows from the triangle inequality before squaring each component. Collecting signed cancellations before taking absolute values can only improve this bound. It does not require estimating unknown Gram entries.

Theorem 8.1 (Combined rational defect certificate)**.** *Let $s_\lambda > 0$ be rational weights and let symmetric rational blocks $\widehat{Z}_\lambda$ satisfy $\widehat{Z}_\lambda \succeq -\eta_\lambda I$, $\eta_\lambda \ge 0$. Suppose the physical expression $\widehat{\mathcal{Q}} = \sum_\lambda s_\lambda \mathbb{E}(Y_\lambda^* \widehat{Z}_\lambda Y_\lambda)$ has*

$$\widehat{\mathcal{Q}} = c^{\mathsf{T}}\mu + \varepsilon, \qquad |\varepsilon| \le \epsilon, \qquad |(W\mu)_j| \le \delta_j, \qquad \ell \le \mu \le u.$$

Then for every rational y,

$$-\sum_\lambda s_\lambda \eta_\lambda B_\lambda \le \widehat{\mathcal{Q}} \le h_{[\ell,u]}(c - W^{\mathsf{T}} y) + \sum_j |y_j|\delta_j + \epsilon. \tag{34}$$

Consequently the premises are incompatible whenever the rightmost quantity plus $\sum_\lambda s_\lambda \eta_\lambda B_\lambda$ is negative.

Proof. The block inequality is valid for complex vectors: writing real and imaginary parts gives $Y^* \widehat{Z} Y \ge -\eta \|Y\|^2$. Sum with positive weights, take expectations, and apply (33). Proposition 7.2 bounds $c^{\mathsf{T}}\mu$ from above; adding the bounded residual proves the other inequality. A lower bound exceeding an upper bound is impossible. □

Here $s_\lambda = 5 \cdot 10^{15-e_\lambda}$ recovers the source scale. For normalized blocks, use $s_\lambda = 10^{-e_\lambda}$ and divide all source coefficients by $5 \cdot 10^{15}$ consistently. Setting all defects to zero and all nonconstant intervals to $[-1, 1]$ recovers the source's last scalar implication, conditional on its exact expansion. Sharper intervals and B_λ can make a defect budget less conservative. Neither the source's residual vector nor the matrices needed to compute an improved residual-coordinate bound are inferred from their reported totals. The recovered literal function lists do, however, permit an executed amplitude calculation:

Shape	Functions	e_λ	B_λ from (33)
$(1,1,1,2)$	5	14	11520
$(1,1,3)$	21	13	27216
$(1,2,2)$	11	13	25344
$(1,4)$	27	14	62208
$(2,3)$	22	14	50688
(5)	12	15	172800

The finite signed sums were computed exactly from the source lists. For these lists, collection and the type bounds produce the same result as the naive group-cardinality estimate; no reduction is credited. In normalized scale,

$$\sum_\lambda 10^{-e_\lambda} B_\lambda = \frac{20853}{3125000000000}.$$

Conditional on a verified expansion with the source's stated negative upper bound and zero identity residual, Theorem 8.1 therefore would still contradict feasibility if a common certified block deficit η obeyed

$$\eta < \frac{2007321335237}{1000944000}.$$

This is an exact sufficient defect budget in unscaled block units, not an eigenvalue estimate, not a claim that the blocks have defects, and not verification of the missing residual identity. More generally the actual budgets enter additively as in (34).

8.2 Why Schur pivots are not spectral margins

If $Z = L\,\mathrm{diag}(p_i)L^*$ with all rational $p_i > 0$, it is positive definite. A certified spectral lower bound is, for example,

$$Z \succeq \frac{\min_i p_i}{\|L^{-1}\|_1 \|L^{-1}\|_\infty} I, \tag{35}$$

because $\|L^{-1}\|_2^2 \le \|L^{-1}\|_1 \|L^{-1}\|_\infty$ and $\|L^* x\| \ge \|x\| / \|L^{-1}\|_2$. As a concrete warning, the matrix

$$40001 \begin{pmatrix} M^2+1 & M \\ M & 1 \end{pmatrix}$$

has bottom-up Schur pivots 40001 and 40001. At vector $(1, -M)$ its Rayleigh quotient is $40001/(1+M^2)$, arbitrarily small. Thus the source threshold $p > 40000$ supports exact positivity, not a uniform Euclidean eigenvalue gap. Relative physical-metric checks or explicit inverse factors are required for a robust spectral statement.

9 Cubic Defect Identity

Keep the source's cubic meanings of S_0, S_3, S_7, including the exact identification of $6 + 30S_0$ with a nonnegative physical energy. Define the defects in its three reduced equations by

$$r_1 = S_3 + S_7 - 1/5, \quad r_2 = S_0 + 12S_3 - 2/5, \quad r_3 = 4S_3 + 2S_7 - 3/5. \tag{36}$$

Proposition 9.1 (Exact cubic residual identity)**.** *For arbitrary real* S_0, S_3, S_7*,*

$$6 + 30S_0 = -18 + 360r_1 + 30r_2 - 180r_3. \tag{37}$$

If $6 + 30S_0 \geq 0$ *and* $|r_j| \leq \epsilon_j$*, necessarily* $360\epsilon_1 + 30\epsilon_2 + 180\epsilon_3 \geq 18$*. In particular equal bounds* $\epsilon_j \leq \epsilon < 3/95$ *are impossible.*

Proof. Subtract twice the first equation of (36) from the third: $2S_3 - 1/5 = r_3 - 2r_1$. Hence $S_3 = 1/10 + r_3/2 - r_1$ and $S_0 = -4/5 + r_2 - 6r_3 + 12r_1$. Multiplying by 30 and adding 6 gives the identity, and the triangle inequality proves both consequences. □

These are defects in the already reduced equations. If the known values $S_6 = S_8 = -1/20$, the zero-orbit values, or the character coefficient identities have errors, they must first be propagated into these r_j. The threshold is not a distance from an approximately unbiased family to an exactly unbiased family, and it does not justify rounding away any upstream error.

10 Verification Methods and Evidentiary Boundary

The local author checks use standard-library integer and rational arithmetic, with finite loops and explicit test counts; all four test groups passed. The source code is not replaced by a transcription that is then described as its checksum-matching verifier. The final local evidence ledger distinguishes these categories:

1. Rational checks of the shared-column 5/25 and 20/25 weights, three interval endpoints, the lossless-elimination example, 16 residual transport cases, and 64 cubic defect triples. Compression and contrast positivity are proved analytically in Section 5; these finite tests are not substituted for their proofs.

2. Direct reproduction of all 11 cubic coefficient-count rows, the signed orbit cardinalities $360, 120, 20, 90$, and both 30-shift rows, using finite modular exponent arithmetic with at most $11 \cdot 180 \cdot 2 \cdot 36$ coefficient-accumulation iterations.

3. Exact character checks on $F_6 = (\exp(2\pi i\, jk/6))_{j,k}$ and T, using integer pairs in $\mathbb{Q}[\omega]$. Each matrix was checked at all 720 row permutations: 720 zero-L tests, 1440 zero-product tests, and 21600 order-one orthogonality tests. Both are symmetric, so their transpose orientation adds no new case. This does not quantify over all Hadamard matrices.

4. Literal-only AST parsing of the recovered verifier and bounded enumeration of all supplied Young sums yielded the amplitude table, type/order inventory, and exact normalized defect budget. No source top-level table-generation call was executed by the author.

No finite sampling proves (2). No scalar replay of (15) verifies the 98 block pivots or the 3429-coordinate mixed expansion. A rank count obtained modulo a prime only proves the selected minor's rank; unused equations may have higher rational rank. The exact equation $WP = 0$ and the identity free rows are what validate substitution on all real moment vectors.

10.1 Source verifier contract

The source README asks for Python 3.8 or later, standard library only, assertions enabled, and a wrapper checking the verifier SHA-256

`104a7306205b8544c8f913f6afdb59603f2e04a207c9ac592c383b80eb0e2aa9.`

It expects the two rank-table lines described above and the exact scalar triple in (15) with scale 5000000000000000. The TeX includes `certificate/verify.py`; the execution wrapper is `verification/run.py` and refers to `verification/verify.py`. Their identity, availability, and hash were checked rather than presumed from similar filenames. The initial acquired tree contains neither listing nor wrapper. The pinned public tree contains `verification/run.py` and `verification/verify.py`, but no `build/certificate/verify.py`. Thus a direct pinned TeX build needs a staging step to resolve its listing; the standalone verifier is recoverable. This is a packaging issue, not evidence that its mathematics is false. Importing that verifier would run its unguarded four-call pipeline, so component audits used a checked AST boundary instead.

10.2 Independent bounded audit incorporated

The independent auditor's `_audit/M537-independent-20261007/AUDIT.md`, `REPLAY.md`, and `receipt.json` were read, together with its harness and machine-readable results. Evidence labels M537-F1–F6 record:

- The ten originally acquired files match the pinned public Git blobs; the 30160-byte recovered verifier matches the printed SHA-256.
- All six integer blocks are positive definite. All 98 reverse Schur pivots exceed 40000 exactly, and independent forward and reverse rational LDL factorizations reconstruct every matrix. These statements use all 1021 lower-triangle integers, not approximate eigenvalues.
- All 98 function lists match independent signed-permutation enumeration. Four function-type product cases and 12 exact rational common-column Gram cases passed; independent columns give different values in all 12 test cases.
- Two small elimination cases passed, including a relation hidden modulo the first prime. Neither the actual 897-row pair witness nor the actual 2767-row mixed witness was regenerated.
- The printed coefficient subtraction passed, but neither residual coefficient value was recomputed from the full quadratic form.

The component replay completed with exit zero and empty stderr under a 120-second supervisor bound; its child was collected. Its omitted calls are precisely `pairtable()`, `check_pair()`, `mixtable(11)`, and `gaps(data,Emix)`. No full-verifier expected rank lines were obtained. This independent evidence establishes important local components, not either headline theorem. The author's cubic count checks above supply additional finite evidence not executed by that sidecar or by the standalone wrapper.

10.3 Reduction contract for any future full replay

A complete verification must connect each generated key to the original sampling definition, prove every canonicalization preserves that moment, check every exact rational substitution, reconstruct all six block matrices from the correct lower triangles, establish their positivity, and reproduce both the pair zero vectors and mixed residual coefficients. Every retained residual coordinate must be a named original moment. Then, and only then, the source's analytic chain S1–S8 and mixed contradiction can be combined into an independently checked headline. This report's local theorems have shorter proofs and do not inherit an unproved global conclusion through terminology.

11 Current Primary Literature and Comparison

Public searches used only public mathematical titles and topics. No private corpus text or confidential report draft was included in a query. Primary public literature was checked on the study date.

The Fourier character framework predates M537: Matolcsi, Ruzsa, and Weiner formulated the relevant Hadamard constraints and conjecture [2]. Gribling and Polak's published 2024 work develops a symmetry-reduced noncommutative moment hierarchy for MUBs and numerical low-degree results [3]. It does not independently validate the particular finite commutative character certificate in M537. The shared-column construction here uses classical Gram positivity but a different explicit function family and therefore needs its own sampling proof.

Bandeira, Doppelbauer, and Kunisky prove a degree-at-most-six limitation for their specified single-unitary positive-definite-polynomial approach [4]. It is not a blanket prohibition on certificates using several transitions, higher order identities, or our contrast sector. Conversely, adding such structure is not itself a proof that the MUB problem has been solved. Peyrl and Parrilo's rational SOS work [5] is a classical antecedent for numerical discovery followed by exact rational certification; the rational dual witness here uses the same separation of search from proof without claiming historical novelty.

Matolcsi, Matszangosz, Varga, and Weiner's *Triplets of mutually unbiased bases* has a 2026 journal version and a July 2025 arXiv version [6]. Its Hadamard-cube approach treats triplets and conjectural extra restrictions, a different structural target from excluding a complete seven-basis family. Weiner's theorem [7] bridges the absence of seven to an upper bound of five by completion from six; it does not bridge five to three. These primary references clarify the problem boundary, not an external endorsement of M537's large payload.

11.1 Advantages and limitations of the present representation

The physical Gram formulation fixes the shared random column, all normalizations, and the exact nullspace created by order-two completeness. The bounds in Theorem 6.1 and the lossless polytope are strictly more informative than ignoring those constraints. Rational dual witnesses are independently checkable and do not require trusting an LP optimizer. The contrast sector provides a concrete extension with a three-factor moment interpretation.

The price is additional bookkeeping: original keys and their bounds must survive canonicalization; congruences can increase rational coefficient sizes; the full physical kernel may be larger than the explicitly known one; and larger necessary moment systems need not be representable. No complexity bound, measured speedup, or reduction in the unknown MUB maximum is demonstrated. If the existing six-block certificate already has a large exact gap, a tighter scalar bound may have little practical value unless it permits smaller certificates or robust checks; this requires data and measurement, not just a numerical factor.

12 Future Research and Contribution Assessment

An appropriate next mathematical experiment is to label the exact retained keys by Theorem 6.1, solve the bounded quotient problem, and export a rational dual witness. Independently, applying the pointwise function dictionary to enlarged lists can prevent order-two type duplication; the present lists contain no such duplication. Enlarging the certificate with $D - S \succeq 0$ is another specific possibility, with proof of every new moment key still required. A source-specific robustness study would optimize the displayed B_λ, recompute them for enlarged lists, and compute rational inverse-factor bounds rather than substituting Schur pivots for spectral gaps.

The realized contribution is the proved order-two physical compression, the contrast-Gram extension, sharper intervals attached to original observables, and an exact bounded-elimination/dual-replay account. The cubic defect identity adds an explicit local error budget. Their combination is a notable local refinement and is assessed C3/5 under the study's contribution policy. The rating does not credit the inherited source headline, an unexecuted full verifier, potential computational speedups, or future contrast-sector experiments. This is an internal evidence-based assessment, not external peer review or a historical novelty claim.

A Reproducible Input and Check Record

Study-root-relative author evidence paths are

`_audit/M537-author-inputs.json`
`_audit/M537-author-checks.py`
`_audit/M537-author-checks.json`
`_audit/M537-author-review.txt`
`_audit/M537-author-qa.json`
`_audit/M537-author-freeze-manifest.json`.

The input ledger records SHA-256 hashes and actual read scope, including the absent included listing. The check record identifies exact bounded tests and their outcomes. The final freeze manifest binds the final TeX, PDF, check script and output, assessment record, and relevant input hashes. Parent review is targeted internal mathematical review, not a verification of every source theorem. PDF generation and visual layout inspection establish artifact integrity, not mathematical truth.

References

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